Speech Understanding

Question 2: Paper Implementation:

Disentangled Representation Learning for Environment-agnostic Speaker Recognition

Abstract

The original work proposes separating speaker identity and environmental factors in learned embeddings to improve robustness across acoustic conditions. Due to computational and dataset constraints, this study implements a reduced reproduction using the LibriSpeech **train-clean-100 (6.3GB)** dataset, which contains approximately 100 hours of clean speech. Synthetic environmental variations are introduced to emulate real-world conditions. The reproduced model is compared against a strong baseline system, and an additional improvement based on supervised contrastive loss is evaluated. Results indicate that while disentanglement does not consistently outperform the baseline in this small-scale setup, the proposed improvement shows gains under certain conditions, supporting the importance of stronger identity clustering in limited-data scenarios.

1. Introduction

Speaker recognition systems aim to identify or verify speakers from audio signals. However, performance often degrades under varying environmental conditions such as noise, reverberation, and channel distortions. The reviewed paper addresses this issue by learning disentangled representations that separate speaker identity from environmental factors.

This report provides a critical analysis of the proposed method and implements a reduced reproduction due to dataset and computational limitations. The goal is to evaluate whether disentanglement improves robustness and to explore potential enhancements under constrained settings.

2. Paper Summary

The paper proposes a framework that decomposes speaker embeddings into two components:

- **Speaker code**: captures identity information
- **Environment code**: captures environmental characteristics

The model is trained using multiple objectives:

- Speaker classification loss
- Environment classification loss
- Adversarial loss to remove environment information from speaker code
- Reconstruction loss

- Decorrelation regularization
- Cross-environment invariance loss

This design aims to produce embeddings that are invariant to environmental changes while preserving speaker identity.

3. Technical Critical Review

Problem

The paper addresses the challenge of environment-induced variability in speaker recognition, which is a well-known limitation in real-world deployments.

Method

The disentanglement approach is theoretically well-motivated and combines multiple loss functions to enforce separation between speaker and environment features.

Strengths

- Clear architectural design for disentanglement
- Comprehensive loss formulation
- Strong motivation for environment robustness

Weaknesses

- Heavy reliance on large-scale datasets with diverse environments
- Complex training objectives that may be difficult to balance
- Sensitivity to hyperparameters and augmentation strategies

Assumptions

- Availability of environment labels or proxies
- Sufficient data diversity for disentanglement to emerge

Experimental Validity

While the original paper demonstrates improvements on large datasets, its effectiveness in smaller or cleaner datasets is uncertain. This motivates the reduced reproduction in this study.

4. Reproduction Strategy

A full-scale reproduction using VoxCeleb was not feasible due to resource constraints. Therefore, a **justified reduced reproduction** was implemented using:

- **Dataset**: LibriSpeech train-clean-100.tar.gz (6.3GB, 100 hours of clean speech)
- Synthetic environment generation:
    - Additive noise

    - Reverberation
    - Band-limited distortion

This setup simulates environmental variability while acknowledging the limitations of the dataset.

5. Implementation Details

Baseline System

- Log-mel spectrogram frontend
- TDNN-style encoder
- Attentive statistics pooling
- AAM-Softmax loss
- Outputs normalized speaker embeddings

Paper-style Reproduction

- Uses baseline encoder
- Adds disentangler module
- Splits embeddings into:
    - Speaker code
    - Environment code
- Loss components:
    - Speaker classification
    - Environment classification
    - Adversarial confusion
    - Reconstruction
    - Pearson decorrelation
    - Cross-environment invariance

Improved Model

- Adds **supervised contrastive loss** on speaker_code
- Encourages stronger clustering of same-speaker samples

6. Evaluation Metrics

The following metrics were used:

- Equal Error Rate (EER)
- Minimum Detection Cost Function (minDCF)
- Top-1 Identification Accuracy
- PCA-based embedding visualization

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7. Experimental Results

Quantitative Results Table

Method	Condition	EER	minDCF	Top-1 Accuracy
Baseline	Clean-Clean	0.541667	0.051687	1.000000
	Clean-Noise	0.458333	0.051885	0.250000
	Clean-Reverb	0.500000	0.048958	0.250000
	Clean-Bandlimit	0.539683	0.051885	0.875000
Paper Reproduction	Clean-Clean	0.500000	0.053373	1.000000
	Clean-Noise	0.543651	0.051885	0.125000
	Clean-Reverb	0.579365	0.051885	0.416667
	Clean-Bandlimit	0.500000	0.051885	0.916667
Improved Model	Clean-Clean	0.500000	0.053770	1.000000
	Clean-Noise	0.502976	0.051885	0.208333
	Clean-Reverb	0.520337	0.051885	0.500000
	Clean-Bandlimit	0.481647	0.051885	0.916667

8. Visualizations

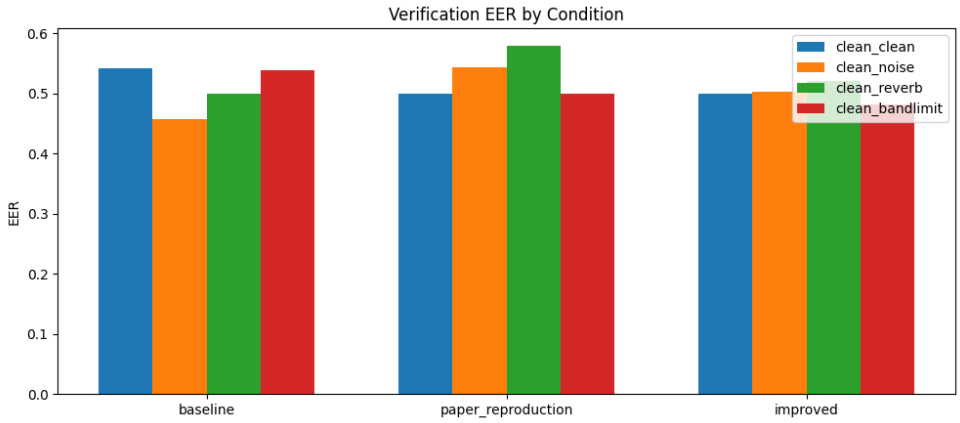

Figure 1: Comparison of EER across different methods and conditions.

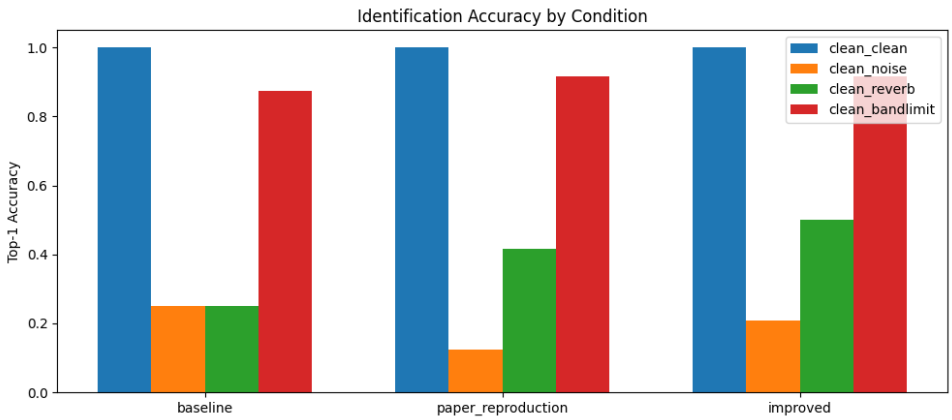

Figure 2: Top-1 identification accuracy across evaluation conditions.

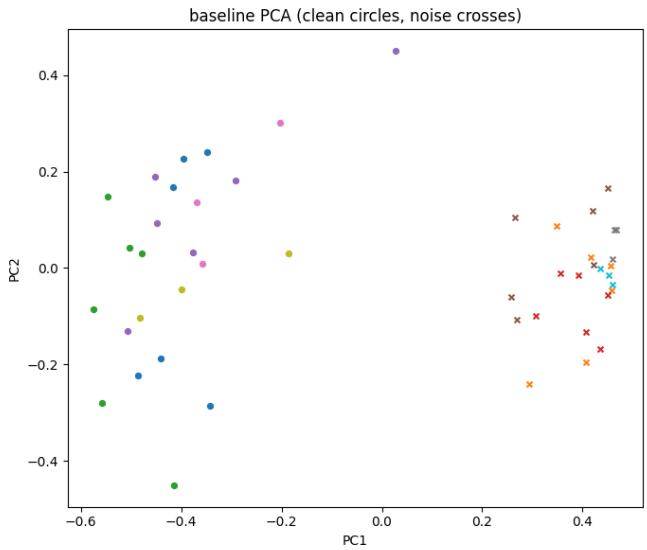

Figure 3: PCA visualization of baseline embeddings.

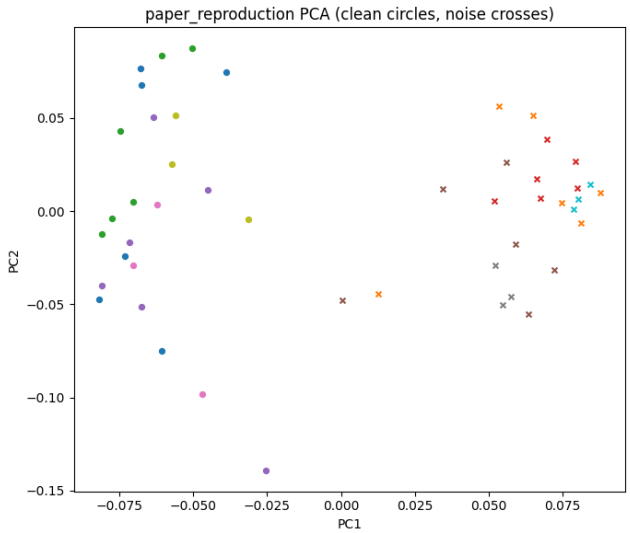

Figure 4: PCA visualization of disentangled embeddings.

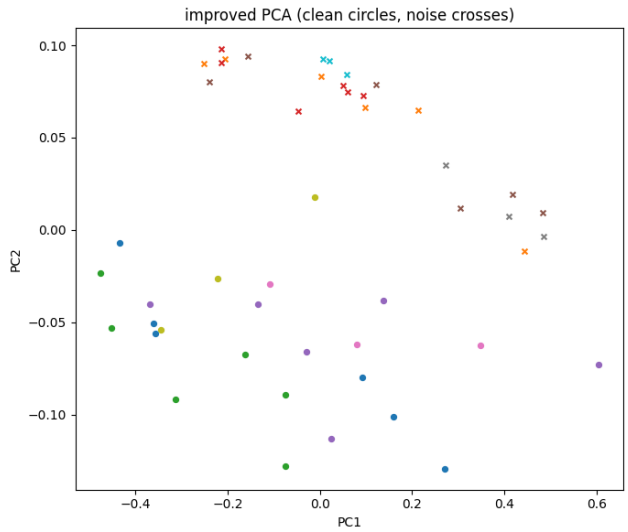

Figure 5: PCA visualization after contrastive enhancement.

9. Improvement Proposal and Analysis

The proposed improvement adds a supervised contrastive loss on the speaker code. The motivation is that the original disentanglement framework does not strongly enforce clustering of same-speaker samples across different environments, especially in a small and clean dataset like LibriSpeech train-clean-100.

The contrastive loss explicitly pulls embeddings of the same speaker closer while pushing different speakers apart. This complements existing losses (adversarial, reconstruction, decorrelation, invariance), which do not directly enforce strong identity clustering.

Results show a mixed but meaningful trend. The improved model performs better than the paper-style reproduction on clean-reverb and clean-bandlimit, indicating improved robustness to structured distortions. However, the baseline performs best on clean-noise EER, suggesting that a simpler model generalizes better to noise in this small-scale setup.

Additionally, the paper-style reproduction does not consistently outperform the baseline, highlighting that disentanglement is sensitive to data scale and augmentation design. Since environments are synthetic, the model may learn augmentation artifacts rather than true real-world variability.

Overall, the improvement partially addresses this limitation by strengthening identity structure. While gains are modest, they support the idea that stronger identity-aware constraints help in reduced-data settings.

10. Discussion

The results show that disentanglement-based methods are highly sensitive to training conditions such as data scale, environment diversity, and optimization stability.

Using LibriSpeech train-clean-100, which is clean and limited in variability, changes the problem compared to the original paper. Environmental differences are synthetically generated, not naturally occurring, which reduces realism.

This explains why gains are limited. The baseline remains strong, while the disentanglement model must balance multiple objectives, which can become unstable with limited data.

The contrastive improvement shows that stronger identity constraints improve robustness, especially for reverberation and band-limiting. However, improvements remain moderate, indicating that optimization alone cannot compensate for limited data.

Overall, disentanglement is promising but context-dependent, and may not outperform simpler models in small-scale settings.

11. Conclusion

This work presented a reduced reproduction of the paper using LibriSpeech train-clean-100 (100 hours clean speech) with synthetic environmental variations.

Results show that:

- The baseline performs best on clean-noise
- The paper-style model does not consistently outperform the baseline
- The improved model performs better on clean-reverb and clean-bandlimit

These findings confirm that disentanglement is sensitive to data scale and training design. The proposed contrastive loss improves identity clustering and provides modest but meaningful gains.

In conclusion, disentanglement is a promising approach, but its effectiveness depends on sufficient data, realistic environments, and strong identity constraints. Future work should focus on larger datasets and better augmentation strategies to fully realize its potential.

12. References

1. LibriSpeech Dataset
2. AAM-Softmax and TDNN-based speaker recognition literature
3. Original Paper: Disentangled Representation Learning for Environment-agnostic Speaker Recognition